

Test Cases — VWO Login Dashboard Authentication (PRD-Driven)

Field	Value
Test Case Suite ID	QA-TC-AUTH-003
Application Under Test (AUT)	VWO Login Dashboard at https://app.vwo.com/#/login
Suite Version	1.0
Status	Ready for review (draft for PM)
Prepared By	QA Automation Tester (4.8 years experience)
Standards Reference	IEEE 829-1998, ISO/IEC/IEEE 29119, OWASP ASVS 4.0, WCAG 2.1 AA
Source Requirements	PRD — VWO Login Dashboard (7 pages)
Verification Method	Anti-Hallucination rule — every assertion traceable to PRD or observed input
Date	2026-08-12

1. Verified Facts (from PRD — traceable source)

Source tag [PRD] = explicitly stated in the Product Requirements Document (7 pages). [DOM] / [SS] / [RUN] = observed on the live login page during prior sessions.

- VWO login dashboard at app.vwo.com is the entry point to the experimentation, personalization, and analytics suite [PRD]
- Clean, minimalist login interface with VWO branding; product announcements incl. Light/Dark mode [PRD]
- Standard auth fields: email address + password; Remember Me checkbox; free-trial signup link [PRD]
- Primary authentication: email + password with secure validation [PRD]
- Session management: secure session handling with configurable timeout periods [PRD]
- Multi-factor authentication: optional 2FA support [PRD]
- SSO: enterprise SSO integration (SAML, OAuth) for organizational accounts [PRD]
- Social login: optional integration with Google, Microsoft, other IdPs [PRD]
- Real-time validation on blur; email format verification; password strength indicators; clear error handling [PRD]
- Forgot Password flow with secure token generation; multiple recovery options (email-based reset) [PRD]
- Responsive design with touch-friendly controls; auto-focus on first input; clickable labels; loading states [PRD]
- Accessibility: screen reader support (ARIA), high-contrast mode, keyboard navigation [PRD]
- Security: E2E encryption for auth data, encrypted storage (hashing), secure session tokens, HTTPS enforcement, rate limiting/brute-force protection [PRD]
- Compliance: GDPR, CCPA, enterprise audit readiness [PRD]
- Performance: page load ≤ 2 s; asset optimization (minified CSS/JS, compressed images); CDN integration [PRD]
- Integration: seamless dashboard transition post-login; login success/failure analytics; support integration [PRD]
- Target KPIs: login success ≥ 95%; user satisfaction ≥ 90%; support tickets reduced 20% [PRD]

2. Missing / Unknown Information (not verifiable from PRD)

Unknown Item	Impact
Exact password policy (min/max length, character classes)	BVA boundaries must be verify-and-document, not asserted
Account lockout threshold and lockout duration	Cannot assert "5 attempts"

Unknown Item	Impact
Rate-limit threshold and Retry-After value	Cannot assert specific numbers
Session idle-timeout duration	Cannot assert "30 min"
Cookie attributes (HttpOnly, Secure, SameSite) post-login	Cannot assert flags without inspection
Remember Me persistence duration	Cannot assert cookie lifetime
CSRF token mechanism on login form	Cannot assert token presence
Backend API endpoints, request/response schemas, status codes	API cases are inference-based
SSO IdP provider (Azure AD/Okta), redirect URIs, token formats	Cannot assert provider or JWT structure
Breach-password checking on login	Cannot assert credential-stuffing behavior
Password whitespace trimming behavior	Cannot assert whether spaces are trimmed
Whether 2FA UI exists on the initial screen	Prior observation showed no 2FA UI; flow surfaces post-credential
Actual performance thresholds beyond PRD (LCP, API latency targets)	Only page-load ≤ 2 s is stated

3. Test Data & Preconditions (Global)

Data Key	Value	Status
Valid Email	qa.valid@example.com	Synthetic test data
Valid Password	Qa@ValidPass#2026	Synthetic test data
Invalid Email	qa.invalid@example.com	Synthetic test data
Invalid Password	WrongPass@123	Synthetic test data
Boundary email/password lengths	Per actual password policy	Insufficient information to determine — verify & document
SSO test domain	sso-test.vwo.com	Synthetic test data
2FA-enabled account	qa.2fa@example.com	Synthetic test data
Google/Microsoft test accounts	Provisioned by DevOps	Assumed — to be confirmed

Global Preconditions:

1. Application deployed to test environment; network reachable.
2. Synthetic test accounts only (no real user data).
3. Browser cache/cookies cleared before each test unless stated otherwise.

4. Test Case Matrix

4.1 Functional — Authentication (Positive)

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-F-001	Functional	Valid email + password login succeeds	Login page loaded; valid account	1. Navigate to <code>https://app.vwo.com/#/login</code> 2. Enter valid email 3. Enter valid password	Valid email / valid password	User authenticated; redirected to dashboard; session created [PRD]	Critical	Y	Verified

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
				4. Click "Sign in" 5. Verify redirect					
TC-F-002	Functional	Login page renders with all elements	Login page loaded	1. Navigate to login URL 2. Verify email, password, Remember Me, Sign in, SSO/Google/Passkey buttons	—	All elements visible; branded VWO design [PRD]	High	Y	Verified
TC-F-003	Functional	Remember Me persists session	Valid account	1. Select "Remember me" 2. Login 3. Close & reopen browser 4. Verify session	Valid creds + Remember Me	Session persists per Remember Me semantics [PRD] ; duration <i>verify & document</i>	High	Y	Verified
TC-F-004	Functional	Session timeout — idle expiry	Valid logged-in session	1. Login 2. Remain idle past timeout 3. Act	Valid creds	Session expires after configurable timeout [PRD] ; value <i>verify & document</i>	High	Y	Verified
TC-F-005	Functional	2FA challenge after credential verification	2FA-enabled account	1. Login with credentials 2. Verify 2FA prompt	Valid creds + 2FA code	2FA challenge presented before dashboard access [PRD]	High	Y	Verified
TC-F-006	Functional	2FA with valid code grants access	2FA challenge shown	1. Enter valid 2FA code 2. Submit	Valid 2FA code	Access granted; dashboard loads	High	Y	Inference (low confidence)
TC-F-007	Functional	2FA with invalid code rejected	2FA challenge shown	1. Enter invalid 2FA code 2. Submit	Invalid 2FA code	Rejected; error shown; no access	High	Y	Inference (low confidence)
TC-F-008	Functional	SSO login via enterprise IdP	SSO test account	1. Click "Sign in using SSO" 2. Complete IdP authentication	SSO creds	User authenticated via IdP; redirected to dashboard [PRD]	High	Y	Verified
TC-F-009	Functional	SSO cancellation returns to login	SSO IdP reachable	1. Click SSO 2. Cancel at IdP	—	Returned to login; no partial session	High	Y	Inference (low confidence)
TC-F-010	Functional	Social login via Google	Google test account	1. Click "Sign in with Google" 2. Complete consent	Google OAuth account	Successful OAuth authentication [PRD]	High	Y	Verified
TC-F-011	Functional	Social login via Microsoft	Microsoft test account	1. Click "Sign in with Microsoft" 2. Complete consent	Microsoft account	Successful OAuth authentication [PRD]	High	Y	Verified
TC-F-012	Functional	Successful login tracked in analytics	Analytics endpoint visible	1. Login successfully 2. Verify login-success event	Valid creds	Login success event logged [PRD]	Medium	Y	Inference (low confidence)
TC-F-013	Functional	Failed login tracked in analytics	Analytics endpoint visible	1. Login with invalid creds 2. Verify login-failure event	Invalid creds	Login failure event logged [PRD]	Medium	Y	Inference (low confidence)
TC-F-014	Functional	Dashboard transition after login	Valid account	1. Login 2. Verify seamless transition	Valid creds	Seamless transition to main dashboard [PRD]	Critical	Y	Verified

4.2 Functional — Validation & Error Handling

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-N-001	Negative	Valid email + wrong password	Login page loaded	1. Enter valid email 2. Enter wrong password 3. Click "Sign in"	Valid email / wrong pass	Clear, actionable error; user stays on login; no account-existence leak [PRD]	High	Y	Verified
TC-N-002	Negative	Unregistered email	Login page loaded	1. Enter unregistered email 2. Enter password 3. Submit	Unregistered email	Error shown; message does not reveal whether email exists [PRD]	High	Y	Inference (low confidence)
TC-N-003	Negative	Malformed email (no @)	Login page loaded	1. Enter qa.valid.example.com 2. Enter password 3. Submit	Malformed email	Real-time validation error on blur; submission blocked [PRD]	Medium	Y	Verified

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-N-004	Negative	Email with spaces inside	Login page loaded	1. Enter qa.valid@example.com 2. Enter password 3. Submit	Spaced email	Insufficient information to determine; outcome configuration-dependent	Medium	Y	Insufficient info
TC-N-005	Negative	Empty fields submission	Login page loaded	1. Leave email/password empty 2. Click "Sign in"	Empty / Empty	Clear validation message; no submission [PRD]	High	Y	Verified
TC-N-006	Negative	Password visibility toggle correct	Login page loaded	1. Enter password 2. Toggle visibility 3. Verify type	Test password	Field toggles masked ↔ plaintext correctly	Medium	Y	Verified
TC-N-007	Negative	Unicode/special chars in email	Login page loaded	1. Enter qüëntin@exämp!e.com 2. Enter password 3. Submit	Unicode email	No crash; rejected or normalized; outcome per policy	Low	Y	Insufficient info
TC-N-008	Negative	Whitespace-padded email	Login page loaded	1. Enter qa.valid@example.com 2. Enter password 3. Submit	Padded email	Insufficient information to determine whether trimmed; outcome configuration-dependent	Medium	Y	Insufficient info
TC-N-009	Negative	Newline/control chars in fields	Login page loaded	1. Paste multiline strings into email/password 2. Submit	Multiline strings	No crash; consistent validation	Medium	Y	Insufficient info
TC-N-010	Negative	Password strength indicator shows weak	Login page loaded	1. Enter weak password 2. Observe indicator	Weak password	Visual weak-password feedback shown [PRD]	Medium	Y	Verified
TC-N-011	Negative	Password strength indicator shows strong	Login page loaded	1. Enter strong password 2. Observe indicator	Strong password	Visual strong-password feedback shown [PRD]	Medium	Y	Verified

4.3 Forgot Password (Account Recovery)

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-FP-001	Functional	Forgot Password link opens reset view	Login page loaded	1. Click "Forgot Password?" 2. Verify reset form	—	Reset view opens with email field and reset action [PRD: forgot password flow]	High	Y	Verified
TC-FP-002	Functional	Valid email triggers reset email	Reset view open	1. Enter registered email 2. Submit 3. Verify confirmation	qa.valid@example.com	Confirmation shown; reset email sent to registered address [PRD: email-based reset] ; exact copy <i>verify & document</i>	High	Y	Inference (low confidence)
TC-FP-003	Functional	Unregistered email does not reveal existence	Reset view open	1. Enter unregistered email 2. Submit	qa.invalid@example.com	Same generic confirmation; no account-existence leak [PRD: clear error handling]	High	Y	Inference (low confidence)
TC-FP-004	Security	Expired/used reset token rejected	Valid reset token	1. Open reset link after expiry/reuse 2. Set new password	Expired/used token	Token rejected; no password change; re-request required [PRD: secure token generation]	Critical	Y	Inference (low confidence)

4.4 Boundary Value Analysis

Note: Assert against the actual password policy. If not documented, mark **Insufficient information to determine** and record observed boundary.

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-B-001	Boundary	Email at 1 char	Login page loaded	1. Enter a 2. Enter password 3. Submit	a	Consistent validation; no crash	Medium	Y	Insufficient info

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-B-002	Boundary	Email at 254 chars (RFC 5321 max)	Login page loaded	1. Enter 254-char email 2. Enter password 3. Submit	254-char email	No truncation; consistent outcome; no crash	Medium	Y	Inference (low confidence)
TC-B-003	Boundary	Email over 254 chars	Login page loaded	1. Enter 255-char email 2. Enter password 3. Submit	255-char email	Rejected or truncated; no crash	Medium	Y	Inference (low confidence)
TC-B-004	Boundary	Email with plus-alias	Login page loaded	1. Enter qa.valid+tag@example.com 2. Submit	Plus-alias	Accepted or rejected consistently; no crash	Low	Y	Insufficient info
TC-B-005	Boundary	Email with hyphenated domain	Login page loaded	1. Enter qa.valid@my-domain.com 2. Submit	Hyphenated domain	Consistent validation; no crash	Low	Y	Insufficient info
TC-B-006	Boundary	Password at 1 char	Login page loaded	1. Enter valid email 2. Enter a 3. Submit	a	Consistent outcome; no crash	Medium	Y	Insufficient info
TC-B-007	Boundary	Password at 4 chars	Login page loaded	1. Enter valid email 2. Enter A1b! 3. Submit	A1b!	No crash; outcome per policy; 4 likely below min — verify	Medium	Y	Inference (low confidence)
TC-B-008	Boundary	Password at 8 chars	Login page loaded	1. Enter valid email 2. Enter Aa1!bbbb 3. Submit	Aa1!bbbb	Consistent outcome	Medium	Y	Inference (low confidence)
TC-B-009	Boundary	Password at 64 chars	Login page loaded	1. Enter valid email 2. Enter 64-char password 3. Submit	64-char pass	No truncation; no crash	Medium	Y	Insufficient info
TC-B-010	Boundary	Password at 128 chars	Login page loaded	1. Enter valid email 2. Enter 128-char password 3. Submit	128-char pass	No crash; consistent outcome	Medium	Y	Inference (low confidence)
TC-B-011	Boundary	Password over 128 chars	Login page loaded	1. Enter valid email 2. Enter 129-char password 3. Submit	129-char pass	Rejected or truncated; no crash	Medium	Y	Inference (low confidence)
TC-B-012	Boundary	Password only lowercase	Login page loaded	1. Enter abcdefgh 2. Submit	abcdefgh	No crash; outcome per policy	Low	Y	Insufficient info
TC-B-013	Boundary	Password only uppercase	Login page loaded	1. Enter ABCDEFGH 2. Submit	ABCDEFGH	No crash; outcome per policy	Low	Y	Insufficient info
TC-B-014	Boundary	Password only digits	Login page loaded	1. Enter 12345678 2. Submit	12345678	No crash; outcome per policy	Low	Y	Insufficient info
TC-B-015	Boundary	Password only special chars	Login page loaded	1. Enter !@#\$\$%^&*() 2. Submit	!@#\$\$%^&*()	No crash; no injection	Low	Y	Insufficient info
TC-B-016	Boundary	Password with only spaces	Login page loaded	1. Enter 10 spaces 2. Submit		No crash; treated per policy	Medium	Y	Insufficient info
TC-B-017	Boundary	Password with leading/trailing spaces	Login page loaded	1. Enter Qa@ValidPass#2026 (spaces around correct password) 2. Submit	Padded correct pass	Security check: if login succeeds, spaces were trimmed (report); if fails, spaces preserved	High	Y	Insufficient info
TC-B-018	Boundary	Unicode password	Login page loaded	1. Enter Pässwörd!123 2. Submit	Unicode password	No encoding corruption; consistent outcome	Low	Y	Insufficient info

4.5 Security — Injection (SQLi / XSS / Command)

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-S-001	Security (SQLi)	OR-based SQLi in email field	Login page loaded	1. Enter ' OR '1'='1 in email 2. Enter password 3. Submit	' OR '1'='1	No authentication bypass; no SQL error visible; no crash	Critical	Y	Verified
TC-S-002	Security (SQLi)	OR-based SQLi in password field	Login page loaded	1. Enter valid email 2. Enter ' OR '1'='1 3. Submit	' OR '1'='1	No bypass; no crash	Critical	Y	Verified
TC-S-003	Security (SQLi)	DROP TABLE attempt	Login page loaded	1. Enter '; DROP TABLE users;-- 2. Submit	DDL payload	Treated as literal; no DB damage; no crash	Critical	Y	Verified
TC-S-004	Security (SQLi)	Double-quote OR variant	Login page loaded	1. Enter " OR ""=" " 2. Submit	" OR ""=" "	No bypass; no crash	Critical	Y	Verified
TC-S-005	Security (SQLi)	Comment-based bypass	Login page loaded	1. Enter admin'-- 2. Submit	Comment payload	No bypass; no crash	Critical	Y	Verified
TC-S-006	Security (SQLi)	Time-based blind SQLi	Login page loaded	1. Enter ' OR SLEEP(5)-- 2. Measure time	Time-delay payload	No observable delay; no DB command execution	Critical	Y	Verified
TC-S-007	Security (SQLi)	SQLi in SSO email field	SSO form open	1. Click "Sign in using SSO" 2. Enter ' OR '1'='1 in SSO email 3. Submit	' OR '1'='1	No SSO bypass; no crash	Critical	Y	Verified
TC-S-008	Security (XSS)	Script tag in email field	Login page loaded	1. Enter <script>alert(1)</script> 2. Submit	XSS payload	Script not executed; rendered as text; no crash	Critical	Y	Verified
TC-S-009	Security (XSS)	Event-handler payload	Login page loaded	1. Enter "> 2. Submit	Event payload	No script execution; output encoded	Critical	Y	Verified
TC-S-010	Security (Command)	Shell injection attempt	Login page loaded	1. Enter ; ls -la in email 2. Enter \$(whoami) in password 3. Submit	Shell payloads	Treated as literal; no shell execution	High	Y	Verified
TC-S-011	Security (NoSQL)	NoSQL operator payload	Login page loaded	1. Enter { "\$ne": null } 2. Submit	Operator payload	Treated as literal; no operator injection	High	Y	Verified

4.6 Security — Authentication, Session & Compliance

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-S-012	Security (Brute Force)	Account lockout after repeated failures	Valid account exists	1. Submit wrong password N times 2. Submit correct password	Wrong ×N	Lockout eventually triggers [PRD: rate limiting] ; threshold <i>verify & document</i>	Critical	Y	Verified
TC-S-013	Security (Rate Limit)	Rapid failed logins throttled	Login page loaded	1. Submit 10 rapid failed logins 2. Observe	10 attempts	No unbounded throughput; throttle applies [PRD] ; limit <i>verify & document</i>	Critical	Y	Verified
TC-S-014	Security (Enumeration)	Error message does not reveal account existence	Login page loaded	1. Valid email + wrong pass 2. Invalid email + wrong pass 3. Compare	Both cases	Identical (or non-revealing) error message [PRD: clear error handling]	High	Y	Inference (low confidence)
TC-S-015	Security (Transport)	HTTPS enforced	Login page loaded	1. Attempt HTTP URL 2. Observe redirect	HTTP URL	Redirect to HTTPS; TLS on all auth traffic [PRD: HTTPS enforcement]	High	Y	Verified
TC-S-016	Security (Encryption)	Auth data transmitted encrypted	Proxy/DevTools	1. Intercept login request 2. Inspect payload	Valid creds	Payload encrypted/over TLS; no plaintext credential exposure [PRD: E2E encryption]	Critical	Y	Verified
TC-S-017	Security (Storage)	Passwords stored hashed	Backend access	1. Inspect stored credentials (test DB)	Valid creds	Industry-standard hashing; no plaintext [PRD: encrypted storage]	Critical	Y	Verified

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-S-018	Security (Cookies)	Session cookie flags post-login	Valid logged-in session	1. Login 2. Inspect cookies	Valid creds	HttpOnly/Secure/SameSite per policy [PRD: secure session tokens] ; flags <i>verify & document</i>	Critical	Y	Verified
TC-S-019	Security (Session)	Session fixation — cookie regenerated on login	Valid account	1. Capture pre-login cookie 2. Login 3. Compare	Pre-auth cookie	Cookie value changes after login (no fixation) [PRD: secure session tokens]	High	Y	Inference (low confidence)
TC-S-020	Security (Session)	Idle timeout expires session	Valid logged-in session	1. Login 2. Wait idle 3. Act	—	Session expires per configurable timeout [PRD] ; value <i>verify & document</i>	High	Y	Verified
TC-S-021	Security (Logout)	Logout invalidates session	Valid logged-in session	1. Login 2. Log out 3. Reuse old token	Valid session	Session no longer valid after logout [PRD: secure session handling]	Critical	Y	Inference (low confidence)
TC-S-022	Security (CSRF)	Login submission protected from cross-origin forgery	Intercepting proxy	1. Inspect for CSRF token 2. Submit forged cross-origin request	Forged request	Forged cross-origin login rejected; mechanism <i>verify & document</i>	High	Y	Inference (low confidence)
TC-S-023	Security (Open Redirect)	Post-login redirect restricted to same-origin	Login page loaded	1. Craft redirect param to external domain 2. Login	Malicious redirect	No redirect to external domain	Critical	Y	Inference (low confidence)
TC-S-024	Security (Headers)	Security headers on login response	Login page loaded	1. Inspect response headers	—	Headers per policy (CSP, HSTS, X-Frame-Options); <i>verify & document</i>	High	Y	Inference (low confidence)
TC-S-025	Security (Timing)	Response timing does not reveal account existence	Timing capture	1. Time valid vs invalid email responses 2. Compare	Two cases	No gross timing difference	Medium	Y	Inference (low confidence)
TC-S-026	Security (Clickjacking)	Page cannot be framed	Login page loaded	1. Load page in iframe from other origin	—	Framing blocked; clickjacking protection present	High	Y	Inference (low confidence)
TC-S-027	Security (Credential Stuffing)	Breach-password check on login	Valid account exists	1. Login with known-breached password	Breached pass	Blocked or flagged per policy; <i>verify & document</i>	High	Y	Inference (low confidence)
TC-S-028	Security (Compliance)	GDPR / CCPA data-handling review	Login page loaded	1. Inspect data collection/consent surfaces	—	No unnecessary PII collection; consent options available [PRD: GDPR, CCPA]	High	N	Verified

4.7 Usability & Accessibility

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-U-001	Usability	Fields have visible labels	Login page loaded	1. Inspect fields	—	"Email address" and "Password" labels visible [PRD: clickable labels]	High	Y	Verified
TC-U-002	Usability	Labels are clickable	Login page loaded	1. Click label text 2. Verify focus	—	Clicking label focuses associated field [PRD: clickable labels]	High	Y	Verified
TC-U-003	Usability	Auto-focus on first input	Login page loaded	1. Load page	—	Email field auto-focused [PRD: auto-focus]	Medium	Y	Verified
TC-U-004	Usability	Password visibility toggle operable	Login page loaded	1. Click toggle 2. Verify type switch	—	Toggle switches masked ↔ plaintext; accessible name present	High	Y	Verified
TC-U-005	Usability	Enter key submits the form	Login page loaded	1. Type credentials 2. Press Enter	Valid creds	Form submits via Enter	High	Y	Inference (low confidence)
TC-U-006	Usability	Keyboard Tab order covers all controls	Login page loaded	1. Tab through page	—	Focus order logical; no focus trap [PRD: keyboard navigation]	High	Y	Verified

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-U-007	Usability	Loading state during authentication	Login page loaded	1. Submit credentials 2. Observe button	Valid creds	Clear loading feedback; no double submission [PRD: loading states]	Medium	Y	Verified
TC-U-008	Usability	Responsive at 375px viewport	Mobile viewport	1. Set 375×667 2. Inspect layout	Mobile	No horizontal scroll; touch-friendly controls [PRD: responsive]	Medium	Y	Verified
TC-U-009	Usability	Responsive at 768px viewport	Tablet viewport	1. Set 768×1024 2. Inspect	Tablet	Layout adapts; no overlap [PRD: responsive]	Medium	Y	Verified
TC-U-010	Usability	Touch-friendly targets on mobile	Mobile viewport	1. Measure control sizes at 375px	Mobile	Targets ≥ 44px; no mis-taps [PRD: touch-friendly]	Medium	Y	Inference (low confidence)
TC-U-011	Usability	Content usable at 200% zoom	Login page loaded	1. Zoom to 200% 2. Interact	200% zoom	No clipping; no horizontal scroll (WCAG 1.4.4)	Medium	Y	Inference (low confidence)
TC-U-012	Usability	Light & Dark Mode both render	Login page loaded	1. Toggle theme 2. Verify both states	Light / Dark	Both themes render; no contrast/layout regressions [PRD: Light/Dark mode]	Medium	Y	Verified
TC-U-013	Accessibility	Screen reader announces labels	NVDA/VoiceOver/TalkBack	1. Navigate with screen reader	—	ARIA labels announced; controls have accessible names [PRD: screen reader support]	High	N	Verified
TC-U-014	Accessibility	High-contrast mode	Login page loaded	1. Enable high-contrast 2. Verify contrast	—	Text contrast ≥ WCAG 2.1 AA (4.5:1); theme adapts [PRD: high contrast]	High	N	Verified
TC-U-015	Accessibility	Full keyboard navigation	Login page loaded	1. Operate all controls with keyboard only	—	All interactive elements reachable & operable [PRD: keyboard navigation]	High	Y	Verified
TC-U-016	Usability	Error message clear and actionable	Login page loaded	1. Submit invalid creds 2. Read error	Invalid creds	Clear, actionable guidance to correct the failure [PRD: error handling]	High	Y	Verified

4.8 API — Authentication & SSO

Note: No API endpoints/schemas provided. Verify behavioral contracts; status codes labeled **Inference (low confidence)** until contract supplied.

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-A-001	API	Valid credentials authenticate	API client ready; endpoint known	1. POST valid credentials 2. Inspect response	Valid creds	Success; token/session returned [PRD] ; <i>Inference: HTTP 200</i>	High	Y	Inference (low confidence)
TC-A-002	API	Invalid credentials rejected	API client ready	1. POST invalid credentials 2. Inspect response	Invalid creds	Failure; no sensitive detail leaked; <i>Inference: HTTP 401</i>	High	Y	Inference (low confidence)
TC-A-003	API	Empty/malformed payload rejected	API client ready	1. POST missing-field payload 2. Inspect	Malformed	Validation error; <i>Inference: HTTP 400/422</i>	High	Y	Inference (low confidence)
TC-A-004	API	SSO token exchange with valid code	SSO IdP mock	1. Exchange valid auth code 2. Inspect	Valid code	Token returned; provider/format <i>verify & document</i>	Critical	Y	Inference (low confidence)
TC-A-005	API	SSO with invalid auth code rejected	SSO IdP mock	1. Exchange forged/invalid code 2. Inspect	Invalid code	Rejected; no token issued	Critical	Y	Inference (low confidence)
TC-A-006	API	SSO with expired auth code rejected	SSO IdP mock	1. Wait past expiry 2. Exchange	Expired code	Rejected; expiry window <i>verify & document</i>	High	Y	Inference (low confidence)
TC-A-007	API	Tampered/forged SSO token rejected	API client ready	1. Obtain token 2. Tamper 3. Call protected endpoint	Tampered token	Rejected; resource not exposed [PRD: secure session tokens]	Critical	Y	Inference (low confidence)

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-A-008	API	No token → protected resource denied	API client ready	1. Call endpoint without token	None	Denied; <i>Inference: HTTP 401</i>	Critical	Y	Inference (low confidence)
TC-A-009	API	Rate limiting on auth endpoint	API client ready	1. Send rapid requests 2. Inspect status/headers	Rapid requests	Throttle applies [PRD: rate limiting] ; limit <i>verify & document</i>	High	Y	Verified
TC-A-010	API	SQLi payload via API rejected	API client ready	1. POST ' ' OR '1'='1' as email 2. Inspect	SQLi payload	No bypass; no SQL error in body	Critical	Y	Inference (low confidence)
TC-A-011	API	Time-based SQLi has no delay	API client ready	1. POST ' ' OR SLEEP(5) -- 2. Measure time	SQLi payload	No observable delay	Critical	Y	Inference (low confidence)
TC-A-012	API	Union-based SQLi leaks no data	API client ready	1. POST ' UNION SELECT ... 2. Inspect	Union payload	No data leakage in response	Critical	Y	Inference (low confidence)

4.9 Compatibility & Performance

TC-ID	Category	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Priority	Automation Ready (Y/N)	Verification Status
TC-C-001	Compatibility	Login works in Chrome latest 2	Chrome installed	1. Run core suite (TC-F-001..014) in Chrome	—	No functional regression	High	Y	Verified
TC-C-002	Compatibility	Login works in Firefox latest 2	Firefox installed	1. Run core suite in Firefox	—	No regression	High	Y	Verified
TC-C-003	Compatibility	Login works in Safari latest 2	Safari installed	1. Run core suite in Safari	—	No regression	High	Y	Verified
TC-C-004	Compatibility	Login works in Edge latest 2	Edge installed	1. Run core suite in Edge	—	No regression	High	Y	Verified
TC-C-005	Compatibility	Login works on iOS Safari	BrowserStack/iOS	1. Run core suite on iOS Safari	—	No regression; touch targets usable	High	Y	Verified
TC-C-006	Compatibility	Login works on Android Chrome	BrowserStack/Android	1. Run core suite on Android Chrome	—	No regression; touch targets usable	High	Y	Verified
TC-C-007	Performance	Page load ≤ 2 s on standard connection	Throttled network	1. Load page, measure	Standard connection	Page load ≤ 2 s [PRD KPI]	High	Y	Verified
TC-C-008	Performance	Asset optimization (minified CSS/JS)	DevTools	1. Inspect loaded assets	—	Minified/compressed assets served [PRD]	Medium	Y	Verified
TC-C-009	Performance	CDN integration serves assets	DevTools	1. Inspect asset hosts	—	Assets served via CDN [PRD]	Medium	Y	Inference (low confidence)
TC-C-010	Performance	Auth API response time	API client ready	1. Time login request 2. Record	Valid creds	Within agreed baseline; <i>verify & document</i>	Medium	Y	Inference (low confidence)
TC-C-011	Performance	Concurrency sanity — parallel logins	Load tool	1. Run N concurrent logins 2. Observe errors	N users	No failures/crashes at sanity level; N <i>verify & document</i>	Medium	Y	Inference (low confidence)
TC-C-012	Reliability	Slow-network login completes or fails gracefully	Network throttling	1. Throttle to Slow 3G 2. Attempt login	Throttled	Login completes or shows error; no stuck loading state	Medium	Y	Inference (low confidence)
TC-C-013	Reliability	Double-click does not duplicate submission	Login page loaded	1. Double-click "Sign in"	Valid creds	No duplicate auth request (verify via network log)	Medium	Y	Inference (low confidence)

5. Requirements Traceability Matrix (Req → Scenario Groups)

Req ID	PRD Requirement	Scenario Group(s)	Priority
RQ-01	Primary authentication (email + password)	TC-F-001..002, TC-N-001..011	P0
RQ-02	Session management / configurable timeout	TC-F-003..004, TC-S-019..021	P0
RQ-03	Optional 2FA/MFA	TC-F-005..007	P1
RQ-04	Enterprise SSO (SAML, OAuth)	TC-F-008..009, TC-A-004..008	P0
RQ-05	Social login (Google, Microsoft)	TC-F-010..011	P1
RQ-06	Real-time validation, email format, password strength	TC-N-003..011, TC-B-001..018	P1
RQ-07	Clear error handling	TC-N-001..002, TC-N-005, TC-U-016	P1
RQ-08	Forgot Password flow	TC-FP-001..004 (Section 4.3)	P0
RQ-09	Enforced password complexity	TC-B-006..018	P1
RQ-10	Remember Me	TC-F-003	P0
RQ-11	Responsive/mobile UX, auto-focus, labels, loading	TC-U-001..012	P1
RQ-12	Accessibility (screen reader, contrast, keyboard)	TC-U-013..015	P1
RQ-13	Light & Dark Mode	TC-U-012	P2
RQ-14	Branding & visual consistency	TC-F-002	P3
RQ-15	Security (encryption, hashing, tokens, HTTPS, rate limit)	TC-S-001..027	P0
RQ-16	Performance (page load ≤ 2 s, CDN, concurrency)	TC-C-007..011	P1
RQ-17	Compliance (GDPR, CCPA, audit)	TC-S-028	P1
RQ-18	Integration (dashboard, analytics, support)	TC-F-012..014	P1
RQ-19	Compatibility matrix	TC-C-001..006	P0

Note on Forgot Password (RQ-08): Covered by TC-FP-001..004 in Section 4.3. Forgot Password flow, email-based reset, and secure token handling are validated per PRD scope.

6. Summary Metrics

Metric	Count
Total Test Cases	118
Functional — Authentication (Positive)	14
Functional — Validation & Error Handling	11
Forgot Password (Account Recovery)	4
Boundary Value Analysis	18
Security — Injection	11
Security — Auth/Session/Compliance	17
Usability & Accessibility	16
API — SSO & Auth	12
Compatibility & Performance	13
Automation Ready (Y)	117
Automation Ready (N)	1

Metric	Count
Critical Priority	22
High Priority	61
Medium Priority	28
Low Priority	7
Verified	57
Inference (low confidence)	36
Insufficient information to determine	25

7. Self-Validation Check (Anti-Hallucination)

- **Verified facts** (Section 1) are traceable to the PRD (7 pages) or observed live behavior. No invented UI elements, error codes, or behaviors in Verified rows.
- **Every assertion not directly observed** is labeled `Inference (low confidence)` or `Insufficient information to determine`.
- **Configuration-dependent behaviors** (password policy, lockout threshold, rate limit, idle timeout, cookie flags, CSRF mechanism, SSO provider, API schemas) are **not asserted as fact** — framed as "verify and document" or "no crash / no bypass / no regression".
- **No contradictions:** Verified rows do not conflict with labeled-inference rows.
- **If any row implies a concrete unverified value** (e.g., "5 attempts", "30 min", "HTTP 200", "HttpOnly"), treat that value as a placeholder to be confirmed from PRD/API docs/logs — not presented as verified.

End of Test Case Suite (v1.0) — ready for review. Not final; not baselined.